

Liblouis User's and Programmer's Manual

This manual is for liblouis (version 3.35.0, 1 September 2025), a Braille Translation and Back-Translation Library derived from the Linux screen reader BRLTTY. It is written in C. It is used in screen readers and accessibility tools to translate text to and from braille.

Overview

Main Source Modules:

- **compileTranslationTable.c** – Compiles and manages translation tables. Provides **lou_getTable**, **lou_checkTable**, and **lou_free**.
- **logging.c** – Handles internal logging.
- **lou_translateString.c** – Translation logic.
- **lou_backTranslateString.c** – Back-translation logic.

Two Header Files:

- **liblouis.h** – Public API definitions.
- **louis.h** – Internal use and by liblouisutdml; includes liblouis.h.

Character Handling:

By default, liblouis handles all characters internally as 16-bit unsigned integers. It can be compiled for 32-bit characters as explained below. The meanings of these integers are not hard-coded. Rather they are defined by the character-definition opcodes. However, the standard printable characters, from decimal 32 to 126 are recognized for the purpose of processing the opcodes. Hence, the following definition is included in liblouis.h. It is correct for computers with at least 32-bit processors.

```
typedef unsigned short int widechar
```

Internal Functions (Do Not Use)

- `resolveSubtable()`

- parseChars()
 - compileRule()
 - compilePassOpcode()
 - _lou_getALine()
 - includeFile()
 - compileHyphenation()
 - _lou_newBuffer()
 - _lou_appendString()
 - _lou_import()
 - _lou_freeBuffer()
 - _lou_Buffer type
 - Any function beginning with _
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Public API Functions (Use These)

- lou_translate()
- lou_translateString()
- lou_backTranslate()
- lou_backTranslateString()
- lou_hyphenate()
- lou_getTable(tableList)
- lou_getTableList()
- lou_checkTable(tableList)
- lou_setDataPath(path)
- lou_getDataPath()
- lou_free()
- lou_charSize()
- lou_logFile()
- lou_logPrint()